Guojie Zhong

guojiezhong@nygenome.org New York, NY zhongguojie1998 guojie-zhong-b377a5179

0000-0001-5664-9069

General Information

Full Name: Guojie Zhong

Date of Birth: 18th May 1998

Languages: English, Chinese

Education

Columbia University, Systems Biology, Integrated Program in Cellular, Molecular, and Biomedical Studies — New York, NY, 2019 – 2024

Peking University, Integrated Science, Yuanpei College — Beijing, China, 2015 – 2019

Experience

New York Genome Center, AI Genomics Fellow — New York, NY, July 2026 – present, 4 months
Develop deep learning methods (DNA language models) to predicting non-coding variant effect, prioritize causal variants for diagnosis and therapeutic target discovery.

Ren Lab, New York Genome Center, Postdoctoral Research Associate — New York, NY, Mar 2025 – June 2026, 1 year 4 months
Develop deep learning methods (DNA language models) to predicting non-coding variant effect, prioritize causal variants for diagnosis and therapeutic target discovery.

Insitro, Inc., Data Science & Machine Learning Intern — South SF, CA, June 2024 – Sept 2024, 4 months
Evaluate and improve the in-house drug target discovery pipeline by integration of GWAS, RNA-seq and PharmaProjects data.

Shen Lab, Department of Systems Biology, Columbia University, Graduate Research Assistant — New York, NY, Sept 2020 – Feb 2025, 4 years 6 months

- Develop deep learning method to predict coding variant effect using representations of protein sequence, structure and functions.
- Develop statistical genetics methods to identify disease risk genes for developmental disorders.

Zhang Lab, Biomedical Pioneering Innovation Center (BIOPIC), Peking University, Research Assistant — Beijing, China, Mar 2017 – July 2019, 2 years 5 months

- Develop computational methods to reconstruct cell spatial organization from single-cell RNA sequencing data based on ligand-receptor mediated self-assembly.

Open Source Projects

EpiBRAIN — 2025 – 2026
Sequence-to-function model for predicting cell-type-specific chromatin profiles, gene expression and non-coding variant effects in human brain.

PreMode — 2022 – 2024
PreMode predicts mode-of-action of missense variants by deep graph representation learning of protein sequence and structural context.

RESCVE 2022 – 2022
A deep learning model to represent the modes of action of missense variants.

VBASS 2020 – 2021
A Bayesian Mixture model with learnable prior to integrate single cell gene expression data in rare variants association analysis.

CSOmap 2018 – 2020
Reconstruction of cell spatial organization from single-cell RNA sequencing data based on ligand-receptor mediated self-assembly.

Honors and Awards

Chinese Government Award for Outstanding Self-Financed Students Abroad 2024
Guojie Zhong

Dean's fellowship, Graduate School of Arts and Science, Columbia University 2019
Guojie Zhong

2014-2015 academic year Outstanding Freshman Scholarship, Peking University 2015
Guojie Zhong

The 28th National Olympiad in Chemistry in Provinces, 1st prize, Zhejiang Province, China 2014
Guojie Zhong

Mentorship

- **Di Liu**, summer rotation PhD student, Ren Lab, New York Genome Center (05/2025–08/2025). Assisted in the project of developing deep learning models to predict the pathogenicity of genetic variants.
- **Demi Zhuang**, summer visiting undergraduate student, Shen Lab, Columbia University (06/2023–08/2023). Developed protein family specific benchmarking tasks and datasets for PreMode. Applied PreMode to those datasets and discovered several novel likely G/LoF variants.
- **Andrew Lee**, summer visiting high school student, Shen Lab, Columbia University (06/2021–08/2021). Developed an R app to visualize the scRNA-seq data of autism risk genes in brain cell types from public datasets. Applied it to discovery of novel autism risk genes and identification of disease associated cell types.

Academic Interests

Genomics & Genetics

Deep Learning, Statistical Modeling, and Bioinformatics

Peer Review Service

- PLOS Computational Biology, Molecular Medicine, Scientific Reports.

Other Interests

Hobbies: Fingerstyle guitar, badminton, etc.